Critic Architecture: Block-by-Block Specification

$$D_\phi : [-1,1]^{1\times 64\times 64} \longrightarrow \mathbb{R}$$

Block-by-block reading

The critic D_ϕ is the topological dual of the generator: it consumes a $1 \times 64 \times 64$ image (either a real training tensor or a fake $G_\theta(z)$) and emits a single real-valued score whose difference of expectations over $\mathbb{P}_r$ and $\mathbb{P}_g$ approximates the Wasserstein-1 distance $W_1(\mathbb{P}_r, \mathbb{P}_g)$. The diagram is again read top-to-bottom and uses the same colour convention as the generator companion: *green* blocks denote tensors at the network boundary ($\mathbf{x}$ and $D_\phi(\mathbf{x})$), *blue* blocks denote internal feature maps $\mathbf{h}^{(\ell)}$ that the critic materializes and discards during a forward pass, and *amber* blocks denote the parameterised operators. The side labels record the spatial resolution at each layer; in the critic, that resolution is *halved* at every stage, mirroring the generator's doubling pattern. Below we walk through every block in the order in which the forward pass visits them.

Block 1 — Input image $\mathbf{x} \in [-1,1]^{1\times 64\times 64}$

The input to the critic is a single-channel 64×64 tensor whose per-pixel values are bounded in $[-1, 1]$. During training, this tensor comes from one of two sources: a *real* batch sampled uniformly from the training tensor bank (a per-bucket min–max-normalised recurrence plot persisted by Stage 2 of the pipeline), or a *fake* batch produced by the generator $G_\theta(z)$ for fresh $z \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{128})$. Both occupy the same support $[-1, 1]^{1\times 64\times 64}$ by construction; the critic itself is never told which source produced a given input. That symmetry is what makes the Wasserstein-1 estimator $\mathbb{E}_{\mathbf{x}\sim\mathbb{P}_r}[D_\phi(\mathbf{x})] - \mathbb{E}_{\mathbf{x}\sim\mathbb{P}_g}[D_\phi(\mathbf{x})]$ a meaningful quantity: each expectation is computed against D_ϕ's *undiscriminating* scoring of inputs from the two distributions.

Block 2 — Stage 1 operator $\mathrm{Conv}^{1\to 64}_{k=4,\,s=2,\,p=1} \circ \sigma_{\mathrm{LReLU}}(0.2)$

Stage 1 is the structural dual of the generator's terminal stage: a stride-2 convolution with kernel size 4 and padding 1 that halves the spatial extent ($64 \to 32$) while expanding the channel count from 1 to 64. The kernel tensor $W_1 \in \mathbb{R}^{1\times 64\times 4\times 4}$ sees, at each output position, a 4×4 patch of the input image, and the stride-2 step guarantees that consecutive output cells tile the input with the same kind of evenly-spaced receptive-field overlap that makes the generator's transposed convolutions artefact-free. The activation is a leaky-ReLU with negative slope 0.2: $\sigma_{\mathrm{LReLU}}(x) = \max(0.2\,x,\, x)$. The leakiness matters here because the critic is updated five times per generator step under the locked $n_{\mathrm{critic}} = 5$ schedule, so any dead neurons that emerge during early training would cripple the inner loop's effectiveness. There is, deliberately, *no* batch normalisation in any stage of the critic: batch normalisation introduces cross-sample dependence inside each mini-batch, which invalidates the i.i.d. assumption of the per-sample gradient penalty that enforces D_ϕ's 1-Lipschitz property.

Block 3 — Hidden state $\mathbf{h}^{(1)} \in \mathbb{R}^{64\times 32\times 32}$

This is the critic's first internal feature map. It contains $64 \cdot 32 \cdot 32 = 65\,536$ scalars, organised as 64 spatial "feature pictures" on a 32×32 grid. Compared with the generator's penultimate hidden state $\mathbf{h}^{(4)}$ (also of shape $64 \times 32 \times 32$), this layer plays the dual role: where the generator was about to commit to high-frequency texture, the critic has just begun extracting it. The 64 channels at this stage already encode early-vision features that any subsequent layer can recombine — edges, blob orientations, intensity gradients — in the recurrence-plot context, the building blocks out of which diagonal lines, vertical laminae, and block clusters will be assembled by deeper stages.

Block 4 — Stage 2 operator $\mathrm{Conv}^{64\to 128}_{k=4,\,s=2,\,p=1} \circ \sigma_{\mathrm{LReLU}}(0.2)$

Stage 2 sets the canonical downsampling pattern that the next two stages will repeat: stride 2, padding 1, kernel 4, channel doubling. Each spatial dimension halves ($32 \to 16$); the channel count doubles ($64 \to 128$). This budgeted trade — spatial expressiveness for channel expressiveness — is the exact inverse of the generator's channel-halving, spatial-doubling pattern, and is what gives the critic-generator pair its overall topological symmetry. The $(4, 2, 1)$ configuration is preferred over $(3, 2, 1)$ for the same checkerboard-avoidance reason that motivated the generator's choice: at stride 2, kernel 4, padding 1, every output cell sees exactly two input cells, with no overlap-induced amplification or attenuation that would introduce systematic spatial bias.

Block 5 — Hidden state $\mathbf{h}^{(2)} \in \mathbb{R}^{128\times16\times16}$

The second internal feature map, with $128 \cdot 16 \cdot 16 = 32\,768$ scalars. This is the first instance of the critic's *progressive-downsampling invariant*: each subsequent stage takes the previous spatial extent and halves it while doubling the channel count, so the total tensor volume stays approximately constant in each hidden layer (here 32 768 vs. 65 536 in $\mathbf{h}^{(1)}$ — actually halving, not constant; the volume contracts gradually until the terminal projection). At this resolution the critic is processing medium-scale recurrence-plot structure such as diagonal bands and rectangular regimes, the very same structures that the generator commits to at its corresponding $\mathbf{h}^{(2)}$.

Block 6 — Stage 3 operator $\mathrm{Conv}^{128\to256}_{k=4,\,s=2,\,p=1} \circ \sigma_{\mathrm{LReLU}}(0.2)$

Structurally identical to Stage 2 except for the channel widths: input 128, output 256. Spatial extent halves once more ($16 \to 8$); channels double ($128 \to 256$). The activation is again leaky-ReLU with the 0.2 negative slope. Note that, unlike the generator, there is no BN between the convolution and the activation. This is not an oversight but a load-bearing design choice: introducing BN at this depth would couple the critic's evaluations of different samples in the same mini-batch, which in turn would render the per-sample gradient penalty $\mathbb{E}_{\hat{\mathbf{x}}}[(\|\nabla_{\hat{\mathbf{x}}} D_\phi(\hat{\mathbf{x}})\|_2 - 1)^2]$ a function of the entire mini-batch rather than a function of each interpolant individually.

Block 7 — Hidden state $\mathbf{h}^{(3)} \in \mathbb{R}^{256\times8\times8}$

Third internal feature map, $256 \cdot 8 \cdot 8 = 16\,384$ scalars. By this depth, the network has compressed the input recurrence plot down to an 8×8 spatial summary encoded in 256 channels. The information-theoretic intuition is that a $1 \times 64 \times 64$ binary-or-near-binary recurrence plot has, by construction, only a modest amount of resolvable structure (its target recurrence rate is 5%, so most pixels carry redundant information), and a $256 \times 8 \times 8$ feature map has more than enough capacity to summarise that structure while remaining compact enough to score globally. The critic's later stages compress this representation further into a scalar.

Block 8 — Stage 4 operator $\mathrm{Conv}^{256\to512}_{k=4,\,s=2,\,p=1} \circ \sigma_{\mathrm{LReLU}}(0.2)$

The penultimate downsampling stage. Channel doubling $256 \to 512$ and spatial halving $8 \to 4$ continue the canonical pattern. This is the last stage that uses the stride-2, padding-1 configuration; the terminal stage that follows is structurally different (kernel 4, stride 1, padding 0) because its job is no longer to downsample but to *collapse* a $4 \times 4 \times 512$ feature volume into a single scalar. Stage 4 leaves behind a feature map whose spatial extent matches the kernel size of the next stage — by design — so the terminal 4×4 convolution receives exactly one position's worth of input at a time and is, in effect, a per-position linear projection.

Block 9 — Hidden state $\mathbf{h}^{(4)} \in \mathbb{R}^{512\times4\times4}$

The fourth and final internal feature map, $512 \cdot 4 \cdot 4 = 8\,192$ scalars. This is the topological mirror of the generator's $\mathbf{h}^{(1)}$ (also $512 \times 4 \times 4$): where the generator *started* building an image from this representation, the critic has just *finished* compressing one. The fact that the two networks meet in the middle at this tensor shape is a coincidence of the locked architecture rather than an enforced constraint, but it is intuitively comforting: the generator and critic operate on the same *latent geometry* at their narrowest point, and the gradient signal that flows backward through the critic informs the generator's design of features at the corresponding depth.

Block 10 — Stage 5 operator $\mathrm{Conv}^{512\to 1}_{k=4,\,s=1,\,p=0}$ (no activation)

The terminal stage of the critic differs from the previous four in three respects: (i) the channel count collapses from 512 to 1, matching the scalar nature of the Wasserstein score; (ii) the kernel size matches the spatial extent of $\mathbf{h}^{(4)}$, so the stride-1 zero-padded convolution collapses the $4 \times 4 \times 512$ volume to a $1 \times 1 \times 1$ scalar in a single step; (iii) there is no activation function on the output. The absence of a final activation is deliberate. A sigmoid would constrain the output to $(0, 1)$, which is fine for the classic GAN binary-classification objective but incompatible with the Wasserstein-1 dual formulation: the Kantorovich–Rubinstein duality requires $D_\phi(\mathbf{x}) \in \mathbb{R}$, not $D_\phi(\mathbf{x}) \in (0, 1)$. A tanh would similarly bound the score and prevent the gradient penalty from being computed against a genuine unbounded scalar field. Leaving the output raw lets the difference of expectations $\mathbb{E}_{\mathbb{P}_r}[D_\phi] - \mathbb{E}_{\mathbb{P}_g}[D_\phi]$ approximate W_1 directly.

Block 11 — Wasserstein score $D_\phi(\mathbf{x}) \in \mathbb{R}$

The output is a single real number per input image, unbounded in either direction. Its absolute scale carries no information — only the *difference* between average scores of real and fake batches matters — so reading individual scores in isolation is not meaningful. The two ways the score is consumed during training are: (i) the Earth-Mover-gap diagnostic, which tracks $\mathbb{E}_{\text{real}}[D_\phi] - \mathbb{E}_{\text{fake}}[D_\phi]$ per critic step and feeds the final-100-step mean and standard deviation into the four-axis Stability Score; (ii) the gradient-penalty loss, which differentiates D_ϕ with respect to interpolants $\hat{\mathbf{x}} = \alpha \mathbf{x}_{\text{real}} + (1 - \alpha) G_\theta(z)$ and penalises departures from $\|\nabla_{\hat{\mathbf{x}}} D_\phi(\hat{\mathbf{x}})\|_2 = 1$. Both consumers use the raw scalar output without any further normalisation, so D_ϕ's final stage is the literal arithmetic dual of the generator's terminal tanh: the generator squashes onto $[-1, 1]$; the critic emits onto $(-\infty, +\infty)$.

Side annotations and arrows

The five labels on the left edge of the pipeline (32×32, 16×16, 8×8, 4×4, 1×1) record the spatial resolution of the hidden state immediately to their right. They make visible at a glance the geometric-progression structure $H^{(\ell)} = H^{(\ell-1)}/2$ for $\ell = 1, \ldots, 4$ that distinguishes Stages 1–4 from the dimensional-collapsing Stage 5. The arrows between blocks denote ordinary forward-pass data flow; at training time the adjoint flow (backpropagation) traverses the same edges in reverse, collecting gradients with respect to every parameter in $\phi = \bigcup_{\ell=1}^{5} \{W_\ell, b_\ell\}$. There are *no* batch-normalisation parameters in this set: the critic deliberately omits BN at every stage to keep each forward pass an independent function of its own input.

Parameter accounting

A complete count of the learnable parameters in D_ϕ illustrates the budget choices implicit in the architecture. The five convolutions contribute, by stage, $1 \cdot 64 \cdot 4 \cdot 4 = 1\,024$, $64 \cdot 128 \cdot 4 \cdot 4 = 131\,072$, $128 \cdot 256 \cdot 4 \cdot 4 = 524\,288$, $256 \cdot 512 \cdot 4 \cdot 4 = 2\,097\,152$, and $512 \cdot 1 \cdot 4 \cdot 4 = 8\,192$ weight scalars respectively (plus a small bias per output channel, omitted here). There are no batch-normalisation parameters because BN is absent throughout. The total is roughly 2.76 million learnable parameters — approximately 73% of the generator's ~ 3.80 million. Stage 4 dominates the budget because both its input and output channel counts are large, the mirror of the generator's Stage 2 dominance. This back-loaded allocation of capacity is intentional: the early critic stages perform geometric reduction whose statistics are largely a function of local image structure, while the deeper stages aggregate that structure into a global verdict and therefore need more capacity. The generator-critic parameter ratio of roughly 1.4 to 1 is a typical W-GAN-GP balance; pushing the critic too small relative to the generator leads to the so-called "mode collapse onto the critic's blind spots" failure mode, while a critic that is too large relative to the generator wastes training budget on a Lipschitz constraint that is already tight.

Why this exact configuration

The critic mirrors the generator's locked DCGAN/W-GAN-GP family, with four deliberate choices recorded here so that they can be revisited as ablations rather than absorbed silently into the spec. **Leaky-ReLU with slope $\alpha = 0.2$:** chosen over ReLU because the critic is updated $n_{\text{critic}} = 5$ times per generator step, so any dead neurons that emerge early in training would cripple the inner optimisation loop; the 0.2 slope follows the original DCGAN choice and the W-GAN-GP reference implementation. **No batch-normalisation in *any* stage:** the per-sample gradient penalty $\mathbb{E}_{\hat{\mathbf{x}}}[(\|\nabla_{\hat{\mathbf{x}}} D_\phi(\hat{\mathbf{x}})\|_2 - 1)^2]$ requires the critic to be a function of *one* input at a time, which BN's cross-sample statistics would break; layer-normalisation would be an acceptable alternative under WGAN-LP, but our locked spec follows the original WGAN-GP paper, which prescribes no normalisation. **No final activation:** the Kantorovich–Rubinstein duality requires $D_\phi(\mathbf{x}) \in \mathbb{R}$, and any squashing function (sigmoid, tanh) would destroy the unboundedness on which the Wasserstein-1 estimator depends. **Kernel** 4 **with stride** 2 **and padding** 1 **for Stages 1–4, and kernel** 4 **with stride** 1 **and padding** 0 **for Stage 5:** the first configuration tiles input receptive fields exactly two-to-one, avoiding checkerboard-like spatial bias; the second configuration is sized so that the receptive field equals the input spatial extent at that depth, which makes the terminal stage a per-position linear projection collapsing $4 \times 4 \times 512 \to 1$. As with the generator, none of these four choices is individually novel; the contribution is the pre-registered combination that lets the four-axis stability diagnostic be defined against a fixed architectural baseline.

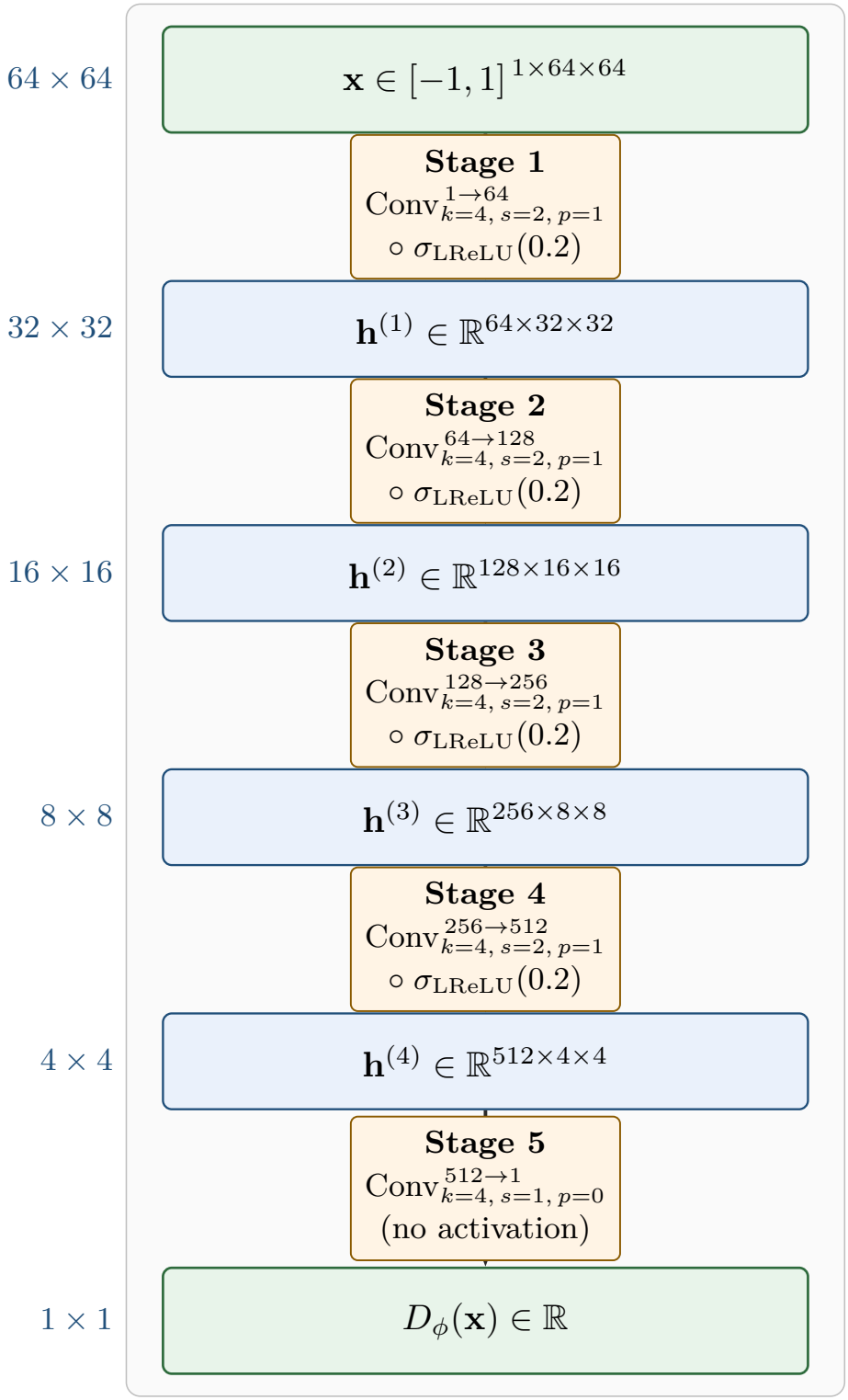


Figure 1: Critic D_ϕ as a deterministic deep convolutional map from a single-channel 64×64 image to an unbounded real-valued score approximating the Wasserstein-1 distance under the Kantorovich–Rubinstein duality.